This project analyzes the Titanic dataset using data cleaning, feature engineering, and visualizations to extract meaningful insights.

```
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
sns.set(style="whitegrid")
```

we import pandas and load the dataset into a DataFrame.

```
# Load dataset
df = pd.read_csv('/content/Titanic-Dataset.csv')

# Show first 5 rows
df.head()
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Cabin	Embarked
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	NaN	S
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C85	C
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	NaN	S
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	C123	S
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	NaN	S

Handle Missing Values

- Filled missing Age values using median
- Filled Embarked using mode
- Dropped Cabin column due to excessive missing values

```
# Check missing values
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	177
SibSp	0
Parch	0
Ticket	0
Fare	0
Cabin	687
Embarked	2

dtype: int64

Data Cleaning

- Filled missing Age values using median
- Filled Embarked using mode
- Dropped Cabin column due to excessive missing values

```
# Fill Age with median
df['Age'] = df['Age'].fillna(df['Age'].median())

# Drop Cabin column
df.drop(columns=['Cabin'], inplace=True, errors='ignore')

# Fill Embarked with mode
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])

# Check again
df.isnull().sum()
```

	0
PassengerId	0
Survived	0
Pclass	0
Name	0
Sex	0
Age	0
SibSp	0
Parch	0
Ticket	0
Fare	0
Embarked	0

dtype: int64

Feature Engineering

We created new useful features:

- FamilySize = SibSp + Parch
- AgeGroup = categorize age into groups

```
# Create FamilySize
df['FamilySize'] = df['SibSp'] + df['Parch']

df[['SibSp', 'Parch', 'FamilySize']].head()
```

	SibSp	Parch	FamilySize
0	1	0	1
1	1	0	1
2	0	0	0
3	1	0	1
4	0	0	0

```
# Create AgeGroup
def age_group(age):
    if age < 12:
        return 'Child'
    elif age < 20:
        return 'Teen'
    elif age < 60:
        return 'Adult'
    else:
        return 'Senior'

df['AgeGroup'] = df['Age'].apply(age_group)

df[['Age', 'AgeGroup']].head()
```

	Age	AgeGroup
0	22.0	Adult
1	38.0	Adult
2	26.0	Adult
3	35.0	Adult
4	35.0	Adult

After preprocessing, we check the final dataset.

```
df.head()
```

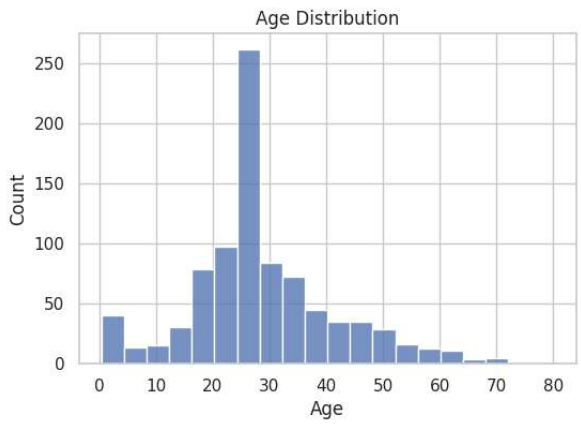
	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare	Embarked	FamilySize	AgeGroup
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500	S	1	Adult
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833	C	1	Adult
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250	S	0	Adult
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000	S	1	Adult
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500	S	0	Adult

1. Histogram – Age Distribution

This chart shows how age values are distributed among passengers.

Most passengers were young adults, with fewer children and elderly people onboard.

```
plt.figure(figsize=(6,4))
sns.histplot(df['Age'], bins=20)
plt.title("Age Distribution")
plt.xlabel("Age")
plt.ylabel("Count")
plt.show()
```

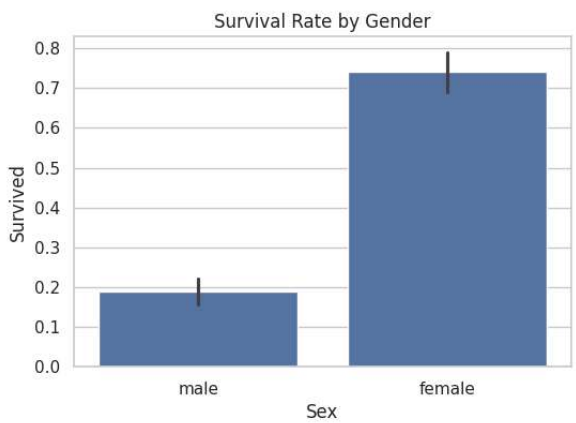


2. Bar Chart - Survival by Gender

This chart compares survival rate between males and females.

Females had a higher survival rate compared to males, indicating priority during rescue.

```
plt.figure(figsize=(6,4))
sns.barplot(x='Sex', y='Survived', data=df)
plt.title("Survival Rate by Gender")
plt.show()
```

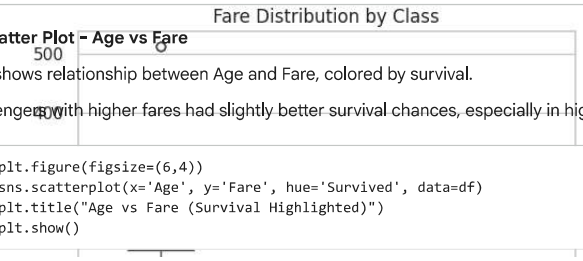


3. Boxplot - Fare by Passenger Class

This shows fare distribution and outliers for each class.

First-class passengers paid higher fares, and there are noticeable outliers with very high ticket prices.

```
plt.figure(figsize=(6,4))
sns.boxplot(x='Pclass', y='Fare', data=df)
plt.title("Fare Distribution by Class")
plt.show()
```

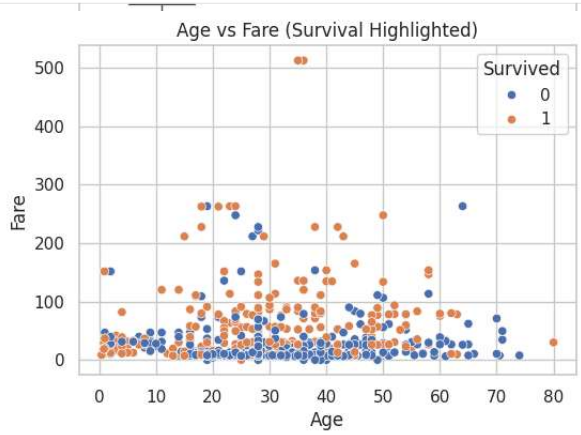


4. Scatter Plot – Age vs Fare

This shows relationship between Age and Fare, colored by survival.

Passengers with higher fares had slightly better survival chances, especially in higher classes.

```
plt.figure(figsize=(6,4))
sns.scatterplot(x='Age', y='Fare', hue='Survived', data=df)
plt.title("Age vs Fare (Survival Highlighted)")
plt.show()
```

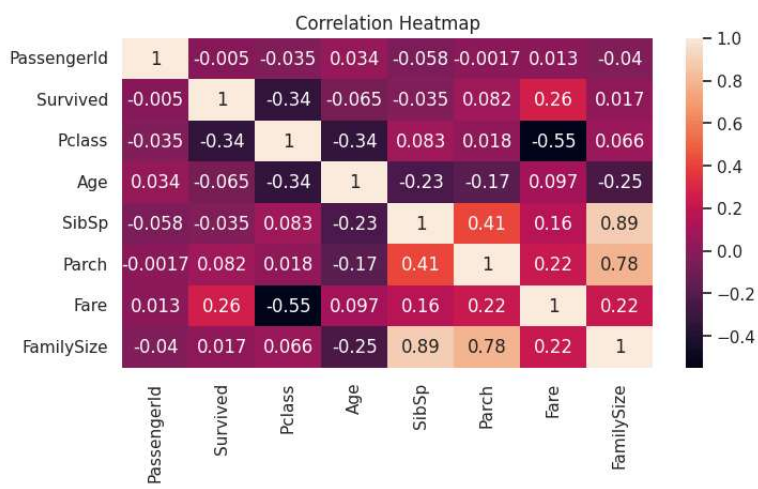


5. Heatmap – Correlation Matrix

This shows correlation between numerical columns.

Fare and survival show a positive correlation, while passenger class has a negative correlation with survival.

```
plt.figure(figsize=(8,4))
corr = df.corr(numeric_only=True)
sns.heatmap(corr, annot=True)
plt.title("Correlation Heatmap")
plt.show()
```



Data analysis shows that gender, passenger class, and fare played an important role in survival. Proper data cleaning and visualization helped uncover these patterns.